Intelligent Multi-Class Text Sentiment Classification Using Machine Learning

Project Report
Muhammad Zain Tariq
BS Computer Engineering

Abstract

This project develops a multi-class sentiment analysis system using NLP, TF-IDF, Logistic Regression, Naive Bayes, and Linear SVM. A Gradio dashboard provides real-time prediction and visualization.

Introduction

Sentiment analysis identifies opinions in text and is widely used for social media monitoring, customer feedback, and review analysis.

Objectives

- Build a sentiment classifier.
- Apply NLP preprocessing.
- Compare three ML models.
- Develop a Gradio interface.

Dataset

Twitter Entity Sentiment Analysis dataset from Kaggle was used. Positive, Negative, and Neutral classes were retained after preprocessing.

	A	B	C	D	E	F	G	H
1	2401	Borderlands	Positive	im getting on borderlands and i will murder you all ,				
2	2401	Borderlands	Positive	I am coming to the borders and I will kill you all,				
3	2401	Borderlands	Positive	im getting on borderlands and i will kill you all,				
4	2401	Borderlands	Positive	im coming on borderlands and i will murder you all,				
5	2401	Borderlands	Positive	im getting on borderlands 2 and i will murder you me all,				
6	2401	Borderlands	Positive	im getting into borderlands and i can murder you all,				

Methodology

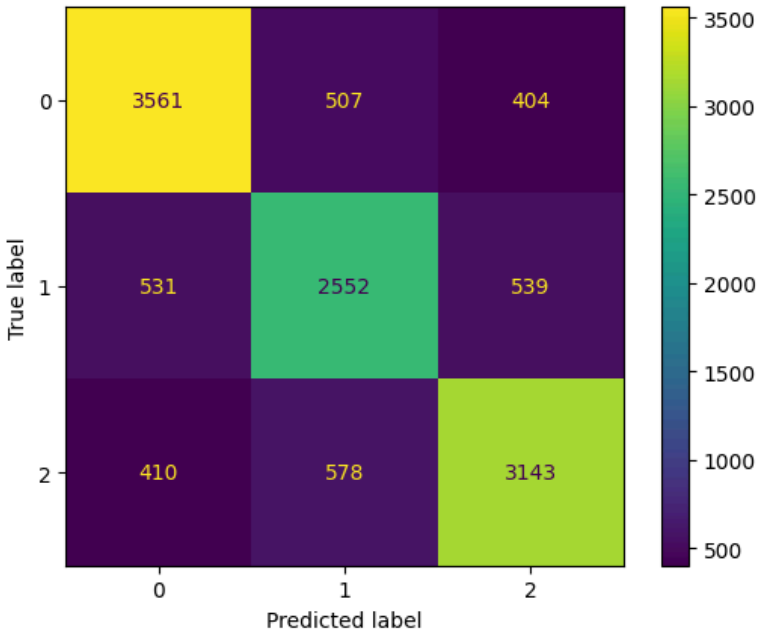
Workflow:
Dataset → Cleaning → Lemmatization → TF-IDF → Train/Test Split → Model Training → Evaluation → Gradio Dashboard

Machine Learning Models

Logistic Regression served as a baseline. Multinomial Naive Bayes provided fast probabilistic classification. Linear SVM achieved the best performance due to its effectiveness on sparse TF-IDF features.

Results

Model Comparison:
- Linear SVM: 0.77
- Logistic Regression: 0.75
- Naive Bayes: 0.71



Gradio User Interface

The dashboard supports text input, prediction, confidence score, prediction history, and sentiment distribution.

4a64108df2049ea262.gradio.live

AI Multi-Class Sentiment Analyzer

TF-IDF + Linear SVM | Twitter Entity Sentiment Dataset

Enter Text

I have interest in AI Cyber And Coud Computing

Analyze

Clear

Prediction

Positive

Confidence Score

39.85

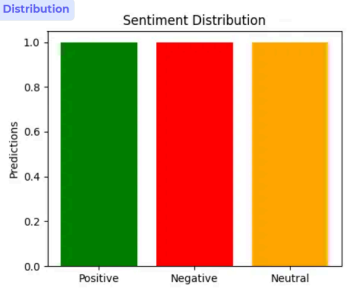
Processing Time (sec)

0.002

Statistics

Total Tested	Positive	Negative	Neutral
3	1	1	1

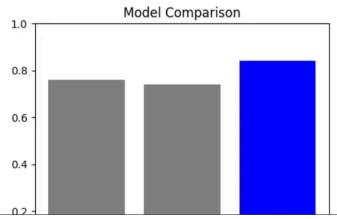
Sentiment Distribution



Prediction History

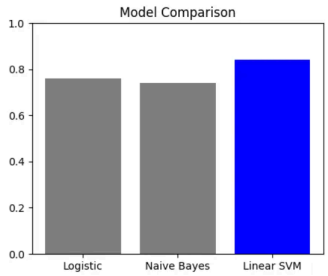
Text	Prediction
I need 20 Marks in subjects	Negative
I have secured A grade in AL Quran Course	Neutral
I have interest in AI Cyber And Coud Computing	Positive

Model Comparison



Activate Windows
Go to Settings to activate Windows.

Model Comparison



Activate Windows
Go to Settings to activate Windows.

Conclusion

The developed system successfully classifies text into Positive, Negative, and Neutral sentiments. Linear SVM produced the best Weighted F1-score, while the Gradio interface makes the solution interactive and user-friendly.

Future Work

- BERT
- RoBERTa
- Hyperparameter tuning
- Cloud deployment